

Week 3 Lecture

Topic: Numbers and Everyday Quantification

Part 1: Listening and oral expression



- a Imagine there was a power cut in your house. What would you find it most frustrating not to be able to do?
- b  9.9 Listen to Chris Stokel-Walker, a journalist from *The Times*, describing his experiment of living for a day with no power. What part of the day do you think he found the most difficult? On the whole, would you say it was a positive or a negative experience for him?
- c Listen again. What things couldn't Chris do during his day without power? Make notes for each of these times.
- 1 7 a.m. – 9 a.m.
 - 2 9 a.m. – midday
 - 3 midday – 2 p.m.
 - 4 2 p.m. – 6 p.m.
 - 5 6 p.m. onwards

d Read some of the expressions Chris uses. What do the **bold** verb phrases mean?

- 1 I read the news on my tablet and the battery life **drops from 65% to 55%** in a few minutes.
- 2 Skype calls really **drain the power**.
- 3 I **close down** unnecessary programs and turn the brightness of my screen down to its lowest level.
- 4 Finally, my **laptop battery dies** just after lunch.
- 5 Thankfully, I can continue to **work in the cloud** on my phone.
- 6 Two and a half years ago, he decided to **live off-grid**.

e Discuss the questions.

- 1 Are power cuts frequent where you live? What causes them?
- 2 Think of a time when you've been without power at home. What did you find difficult? Was there anything you liked about not having power?
- 3 Do you ever suffer from 'battery anxiety'? How low does the battery need to go for you to start worrying?
- 4 What do you think about people like Chris's friend Joel, who live off-grid? Is it good to live without technology?

Part 2: Grammar – Quantifiers

a With a partner, read sentences 1–6 and think about what the missing words could be. Don't write them in yet.

- 1 I used to have _____ of different gadgets, but now I use my phone for almost everything.
- 2 I'd like to have a better computer, but I don't have _____ to buy one right now.
- 3 I spend _____ time online. I think I need to take a break.
- 4 I have a lot of friends on social media, but only _____ of them are close friends.
- 5 I never watch TV or films on my phone, because the screen isn't _____.
- 6 I like Apple products, but I can't afford them – I think they're _____.

b  **9.10** Now listen and complete the sentences in **a**. Did you guess correctly?

quantifiers

large quantities

- 1 My daughter has **a lot of apps** on her phone.
Nina has **lots of clothes**.
I've been there **loads of times**.
- 2 James eats **a lot**.
- 3 There aren't **many cafés** near here.
Do you have **many** close **friends**?
Do you watch **much** TV?
I don't eat **much chocolate**.
- 4 Don't run. We have **plenty of time**.

9.11

- 1 We use *a lot of* or *lots of* in $\boxed{+}$ sentences. We can also use *loads of*, but it's more informal.
- 2 We use *a lot* when there is no noun, e.g. *He talks a lot*. **NOT** *He talks a lot of*.
- 3 *much* / *many* are normally used in $\boxed{-}$ sentences and $\boxed{?}$, but *a lot of* can also be used.
- 4 We use *plenty of* in $\boxed{+}$ sentences. (= more than enough)

small quantities

- 1 **A** Do you want some more ice cream?
B Just **a little**.
The town only has **a few cinemas**.
- 2 I'm so busy that I have **very little time** for myself.
Sarah isn't popular and she has **very few friends**.
- 3 I have **less free time** than I used to have.
There are **fewer flights** in the winter than in the summer.

9.12

- 1 We use *little* + uncountable nouns, *few* + plural countable nouns.
• *a little* and *a few* = some, but not a lot.
- 2 *very little* and *very few* = *not much* / *many*.
- 3 The comparative of *little* is *less* and the comparative of *few* is *fewer*.

more or less than you need or want

- 1 I don't like this city. It's **too big** and it's **too noisy**.
You're speaking **too quietly** – I can't hear you.
- 2 There's **too much traffic** and **too much noise**.
There are **too many tourists** and **too many cars**.
- 3 There aren't **enough parks** and there aren't **enough trees**.
The buses aren't **frequent enough**.
The buses don't run **frequently enough**.

9.13

- 1 We use *too* + adjective or adverb.
- 2 We use *too much* + uncountable nouns and *too many* + plural countable nouns.
- 3 We use (not) *enough* before a noun, e.g. (not) *enough eggs* / *milk*, and after an adjective, e.g. *It isn't big enough*, or an adverb, e.g. *You aren't walking fast enough*.

zero quantity

- 1 There **isn't any milk** in the fridge.
We **don't have any eggs**.
- 2 There's **no milk** in the fridge. We **have no eggs**.
- 3 **A** How many eggs do we have?
B **None**. I've used them all.

9.14

- 1 We use *any* + uncountable or plural noun for zero quantity with a $\boxed{-}$ verb.
- 2 We use *no* + uncountable or plural noun with a $\boxed{+}$ verb.
- 3 We use *none* (without a noun) in short answers.

Activity 1

Circle the correct word or phrase. Tick (✓) if both are possible.

My husband has *too much* / *too many* gadgets.

- 1 I just have to reply to *a few* / *a little* emails.
- 2 Do you spend *much* / *many* time on social media?
- 3 My bedroom is a nice size. There's *enough room* / *plenty of room* for a desk.
- 4 I know *very few* / *very little* people who speak two foreign languages.
- 5 My brother has downloaded *a lot of* / *lots of* apps onto his new phone.
- 6 I have some cash on me, but not *a lot* / *a lot of*.
- 7 Their new TV is *too* / *too much* big. It hardly fits in the living room.
- 8 *There aren't any* / *There are no* potatoes. I forgot to buy some.
- 9 My niece isn't *old enough* / *enough old* to play with a games console.
- 10 I don't have *a lot of* / *many* close friends.

Activity 2

Are the **highlighted** phrases right (✓) or wrong (X)?
Correct the wrong ones.

My nephew got **lots of video games** for his birthday. ✓
I don't post **much photos** online. *many photos*

- 1 'How many presents did you get?' 'A lot of!'
- 2 I buy **fewer e-books** than I used to because I prefer physical books.
- 3 **There isn't no time** to walk there. We'll have to get a taxi.
- 4 Please turn that music down. It's **too much loud**!
- 5 **There aren't many good programmes** on TV tonight.
- 6 My broadband **isn't enough fast** for me to download films easily.
- 7 I get **too much emails** at work. It takes me ages to read them all!
- 8 **A** How much fruit do we have? **B** **Any**. Can you buy some?
- 9 There are **only a little people** that I can talk to about my problems.
- 10 Katarina has **plenty of money**, so she always has the latest phone.

Part 3: Reading and oral expression

a Work with a partner and answer the questions. Who has a more organized digital life?

- Approximately how many emails do you have in your inbox? Do you organize them into folders?
- How often do you add new apps on your phone or delete old ones?
- How many photos do you have on your phone? Are they organized into albums? How quickly could you find a photo you wanted to show someone?
- How many friends do you have on social media? How many of them are real friends? Do you ever unfollow your friends?
- How many people in your 'contacts' have you not contacted in the last year?
- How many different passwords do you have? How do you remember them?
- Do you accept cookies when you visit new websites? Why (not)? How often do you unsubscribe from marketing emails?
- What do you do with your old digital devices, e.g. old phones, tablets, etc.?

b Look at the title of the article and the headings. Tick (✓) the areas of your digital life that you think you need to tidy up.

c Read the article and complete the gaps with a verb from the list.

answer click find forget free up
keep recycle update

d Search the text. Find six words with the prefix *un-* to complete the extracts.

Tip 1 Why should you have hundreds of un_____ or un_____ emails?

Tip 2 Un_____ software or apps that you don't use.

Tip 4 You can 'un_____' friends without them knowing...

Tip 7 If you get un_____ emails from companies and organizations, don't just delete them – un_____.



8 ways to tidy up your digital life

A clean, tidy room makes you feel better about your home. In the same way, a tidy phone, tablet, or computer makes you feel better about your digital life. So if your digital life is a mess, try these tips – the benefits are huge.

1 Inbox messages

Most people have too many emails in their inbox. You don't have hundreds of unopened or unanswered letters in your house, so why should you have hundreds of unopened or unanswered emails? If you can _____ an email in less than two minutes, do it right away. If it will take longer, don't leave it in your inbox – move it into a 'work in progress' folder and reply later.

2 Old software or apps

Uninstall software or apps that you don't use. This will _____ a lot more space on your hard drive or phone.

3 Photos

You wouldn't put bad photos in a physical photo album, so don't keep bad photographs (or videos) on your phone – just delete them. Having poor-quality photos just makes it more difficult to _____ a good photo when you need one.

4 Friends

Having too many friends on social media makes it more difficult to _____ in touch with the ones you really care about. You can 'unfollow' friends without them knowing, so you won't hurt their feelings.

5 Contact information

_____ contact information regularly and delete contacts you no longer need. Most people don't do this often enough.

6 Passwords

Use a password manager app, like 1Password. This gives you as many different passwords as you need and remembers them for you. You'll never _____ a password again.

7 Email marketing

If you get unwanted emails from companies and organizations, don't just delete them – unsubscribe. It should only take a few seconds. Just _____ on the 'unsubscribe' link at the bottom of the email.

8 Old devices

If you've been using technology for any length of time, you probably have a collection of things that you no longer use – memory sticks, USB cables, chargers, old phones, and tablets. If you can't give them to somebody who would use them, _____ old devices properly.

- e Complete the sentences with *un-* and the words from the list.

clear comfortable do follow
helpful known lock read

- 1 Why do you have over 100 _____ emails?
 - 2 If you want to _____ what you've done, press Ctrl+Z.
 - 3 I had to _____ her as she was posting some really annoying things.
 - 4 I can't _____ my phone – I've forgotten the password.
 - 5 The IT Support person was very _____. I still can't print anything.
 - 6 I didn't answer the phone, because it said 'caller _____'.
 - 7 I can't set up the new router – the instructions are really _____.
 - 8 I hate earbuds – I find them really _____.
- f Which of the tips in the article would be most useful for you? Choose your top three. Then discuss your choices in small groups and say why.

Part 4: Numbers - Practice

A. Types of numbers

A **series** or **set** of numbers is a group of numbers together.

1, 3, 5, 7 ... = **odd numbers**; 2, 4, 6, 8 ... = **even numbers**; 2, 3, 5, 7, 11 ... = **prime numbers** [numbers that cannot be divided by any other number, apart from themselves and 1.]

An **approximate number** is one which is roughly correct but is not the **precise** or **exact number**. Work out the approximate answer in your head, then use a calculator to find the exact number.

A **discrete** number or unit is something which is separate and cannot be divided into smaller numbers or units of the same thing.

An **aggregate** is a number reached by totaling a set of numbers = the **total**. The **average** exam mark is calculated by taking the **aggregate** of all the marks and dividing by the number of exam entries.

A **random** number is one chosen by chance, i.e. you cannot predict it.

B. Working with numbers

The word **figure** is often used to refer to the symbol used for a number. *Write the total number in words and figures.*

You can **calculate**¹ a number, **estimate**² a number, round a number **up/down**³ and **total**⁴ a set of numbers. Numbers can **tally**⁵. *My figures don't seem to tally with yours.* You can also **deduct**⁶ one number from another number.

Values are individual numbers in a set of data. *The graph shows the temperature values for different months of the year.* **Variables** are characteristics that can take on different values for different members of a group or set being studied. *In investigating living standards you must take key variables such as social provision and cost of living into account.*

The **incidence** of something refers to how frequently it occurs. *The incidence of twins in the population is growing.* When talking about numbers, **magnitude** simply refers to the size of something, whereas in other contexts it indicates large size or importance. *Write down the numbers in order of magnitude, beginning with the smallest.*

When **making calculations** in, say, an exam, it is often a good idea to make an **estimate**⁷ first of what the answer is likely to be. Then you will see if your final answer is **in the right area**⁸ or not. Exam candidates are also often advised to show their **workings**⁹ so that the marker can see how they **arrived at** their answer and they may get credit for their method even if the final answer is incorrect.

¹ work out ² make a rough guess at ³ make a **fraction**, e.g. $\frac{1}{6}$ or 0.78 into the nearest **whole number**

⁴ add up ⁵ match, agree ⁶ take away, subtract ⁷ rough guess ⁸ approximately the same

⁹ all the calculations leading up to an answer

Activity 1

Answer the questions.

- 1 What is five squared?
- 2 What is the next prime number after 19?
- 3 How is this sequence of numbers created? 3, 9, 27, 81
- 4 What is the aggregate of this set of test marks? 6, 8, 9, 5, 6, 7
- 5 If you round up 6.66, what number do you have?
- 6 $\frac{7}{8}$ and 4 – which is a whole number and which is a fraction?
- 7 In your country is tax automatically deducted from employees' earnings?
- 8 Is an accountant pleased or displeased if figures that he/she is checking tally?

Activity 2

Dr Syal is advising one of his dissertation students who is interested in pollution in road tunnels. Complete the conversation with the correct words.

Dr Syal: You could ¹c..... the total number of private cars that use the tunnel each week, based on the day-to-day figures, and get an ²a..... figure for how much carbon they're all emitting.

Melissa: How ³p..... would that figure have to be?

Dr Syal: Oh, it doesn't have to be exact, you just need to ⁴e..... more or less what the total pollution will be. Then you can check to see if those figures ⁵t..... with the figures that have already been published for similar tunnels. And the figure won't be ⁶c..... of course; it'll go up and down depending on lots of factors such as weather conditions, average speed, etc.

Melissa: But can we say if the figures will be true for the future too?

Dr Syal: Well, we do know that traffic has been growing over the past ten years; it hasn't ever gone down, so I think you can make some useful predictions.

Melissa: Should I present each daily total as a ⁷d..... item or can I just put them all together into one figure for each week?

Dr Syal: A weekly total is fine, and you can ⁸..... it up or ⁹..... to the nearest 100.

Melissa: Right, OK. Thanks so much for your help.

Activity 3

Rewrite the sentences using the correct form of the word in brackets.

- 1 There were fewer accidents last year. (INCIDENCE) *The incidence of accidents declined last year.*
- 2 We made a rough guess at what the final figure might be. (ESTIMATE)
- 3 The graph shows the results from the lowest to the highest. (MAGNITUDE)
- 4 A computer program helped us work out the significance of the different variables. (CALCULATE)
- 5 Taking x away from y will help you arrive at the correct answer. (SUBTRACT)
- 6 The results from the first experiment were not the same as those we got from the repeat experiment. (TALLY)

Activity 4

Complete the email from a maths tutor using words from the box. There is one word you don't need. There is also one deliberate mistake in the email. Can you find it and correct it?

calculations variables
figures reached
area workings
values arrived

Hi everyone,
Just a quick note before you take the exam. Things to remember:
The exam lasts one and a half hour, so plan your time carefully.
Don't forget to show all your ¹..... as we want to see how you
²..... at your results. Take great care when you make your ³.....
– you'd be amazed at how many people submit answers that are hardly
even in the right ⁴..... . And please write legibly – we must be able to
distinguish all your ⁵..... . When doing graphs, plot the ⁶.....
carefully and if asked to describe an experiment don't forget to take all
significant ⁷..... into account. Good luck!
Helen Ward

C. Talking about numbers

Describing proportions

A The study found that the **majority** (56%) of those questioned were non-smokers. These **statistics** show that the number of smokers varies significantly by city. The highest **percentage** of smokers was found in Naples (49.5%).

B According to one study of Italian smoking habits, Naples has the **highest proportion** of smokers (Ficarra *et al.* 2011).

C Today only two **per cent** of the US population works on farms, while another three per cent works in the food processing and food service industries. This small **fraction** of the population (five per cent) is able to grow, process and serve enough food to meet the needs of the whole country.

D Perhaps surprisingly, only a small **minority** of US workers (around 5%) are employed in food production, yet they produce enough food for the whole population (Bowles *et al.* 2005).

1 Match the phrases from the academic texts, A and C, to phrases with the same meaning from the student citations, B and D.

- 1 the highest percentage of smokers _____
- 2 this small fraction of the population _____

2 Do the underlined phrases refer to a large quantity (L) or a small quantity (S)?

- 1 This type of treatment is appropriate in only a minority of cases.
- 2 Fuel costs make up a tiny fraction of the firm's total operating costs.
- 3 The majority of students at the college speak English as a second language.
- 4 A high proportion of their income is spent on food and accommodation.
- 5 This disease affects a relatively small percentage of the population.

Usage note *per cent*, *percentage* and *proportion*

Use **per cent** or % after a number:

*five **per cent** of the population*

*inflation rose by two **per cent***

Use **percentage** or **proportion** to talk more generally about a quantity:

*the **highest percentage** of smokers*

*only a **small percentage** of students*

*a **high/large proportion** of patients*

*a relatively **small proportion** of total spending*

3 Complete the sentences using the word in CAPITALS. Add any other words needed.

► EXAMPLE: A high percentage of employees (74%) work more than 45 hours. PERCENTAGE

- 1 These examples represent only _____ the total number of cases (less than 4%). FRACTION
- 2 The survey found that _____ customers (82%) are satisfied with the service. MAJORITY
- 3 _____ patients (approximately 10%) suffer more severe symptoms. PROPORTION
- 4 In 1900, only 20 _____ American women worked outside the home. PER CENT
- 5 In _____ people (less than 3%), the problem may continue into adulthood. MINORITY

Talking about averages

E When looking at data, it is useful to have some idea of a typical **value** for the data. This is usually called an **average**. When you know the average for a set of data, you have some idea of what value or values you might expect from other data from the same source. If the data is from a sample, you can use the average of the sample to estimate the average for the whole population.

When people talk about averages they are usually talking about the **mean**. The mean is obtained by **calculating** the **total** of all the data values and then **dividing by** the number of data values. This can be shown in a **formula**:

$$\text{mean} = \frac{\text{total of all the data values}}{\text{number of data values}}$$

Source: Adapted from Procter-Green & Winters, (2009), *AQA Statistics GCSE*. Cheltenham: Nelson Thornes.

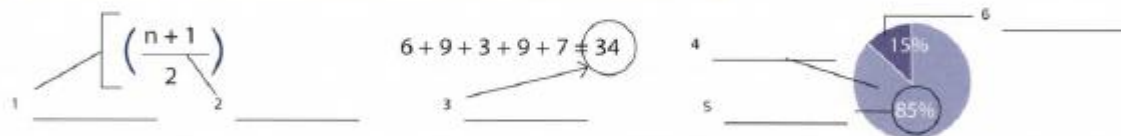
4 Read text E and complete the statements using words from the box.

average calculate divide formula mean total value

- You can use an _____ to represent a typical _____ for a set of data.
- The most common way of measuring an average is called the _____.
- You can _____ the mean using the _____ shown in the text.
- You work out the _____ of all the values, then you _____ that by the number of values in the set.

5 Label the pictures using words from the box.

divide by formula majority minority percentage total



6 Complete the text describing the table.

If we look at the table showing the weather conditions for the island of Madeira, we can see that October to March are the wettest months. To ¹_____ the ²_____ rainfall for this whole period, we first work out the ³_____ of the rainfall ⁴_____ for these six months: $76 + 89 + 84 + 64 + 74 + 79 = 466$. Then we _____ this by the number of months: $466 \div 6 = 77.66$. So 77.66 mm is the ⁶_____ amount of rain during the winter months. Looking at these ⁷_____, we can also say that throughout the year, the ⁸_____ of wet days each month is relatively low. The table also shows that for the ⁹_____ of the year, the maximum temperature is above 19°C and for a large ¹⁰_____ of the year, the temperature does not drop below 15°C .

	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Rain (mm)	64	74	79	33	18	5	0	0	25	76	89	84
Wet days	6	6	7	4	2	0	0	0	3	7	6	7
Max temp	19	18	19	19	21	22	24	24	24	23	22	19
Min temp	13	13	13	14	16	17	19	19	19	18	16	14

Figure 1: Weather conditions in Madeira

